

Failure of the increasing-inclusion claim in Conjecture 00000008586

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October 4, 2026

Abstract

For the Hermitian matrix $T = \text{diag}(0, 1)$, the scalar $1/2$ belongs to the first compression range but not to the second. This disproves the conjectured inclusion $\Lambda_1(T) \subseteq \Lambda_2(T)$. The proof uses the rank-of-compression definition printed in the source. The same matrix and scalar also contradict the proposed direction of inclusion under the usual scalar-compression definition. The accompanying Lean 4 project checks actual isometric compressions, their determinants, and their matrix ranks.

1 The source's definition

Both language versions assert $\Lambda_k(T) \subseteq \Lambda_{k+1}(T)$. They describe membership using the condition that the compression of $\lambda I - T$ have rank at most $k - 1$. Written with an orthonormal frame for a k -dimensional subspace, this is

$$\Lambda_k^{\text{src}}(T) = \{\lambda \in \mathbb{C} : \exists Q \in \mathbb{C}^{n \times k}, \quad Q^*Q = I_k, \quad \text{rank}(Q^*(\lambda I_n - T)Q) \leq k - 1\}. \quad (1)$$

Here Q^* is the conjugate transpose. The equation $Q^*Q = I_k$ says that Q is an isometric embedding of the standard complex Euclidean space $\mathbb{C}^k$ into $\mathbb{C}^n$; the displayed product is its genuine compression. We use (1) for the main disproof.

Proposition 1. *For $T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ and $\lambda = 1/2$, one has $\lambda \in \Lambda_1^{\text{src}}(T)$ but $\lambda \notin \Lambda_2^{\text{src}}(T)$.*

Proof. Take the one-column frame

$$q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad q^*q = 1, \quad q^*Tq = \frac{1}{2}.$$

Thus $q^*((1/2)I_2 - T)q = 0$ has rank zero, proving first-order membership.

For any two-column frame Q with $Q^*Q = I_2$, taking determinants gives $\det(Q^*)\det(Q) = 1$. Put $B = (1/2)I_2 - T = \text{diag}(1/2, -1/2)$. Then

$$\det(Q^*BQ) = \det(Q^*)\det(B)\det(Q) = \det(B) = -\frac{1}{4} \neq 0.$$

Consequently every such compression has rank two, which exceeds $2 - 1 = 1$. This excludes $1/2$ from $\Lambda_2^{\text{src}}(T)$ and proves the failure of the asserted inclusion. $\square$

2 The usual scalar-compression convention

The standard higher-rank numerical range is conventionally defined by scalar compressions [2]:

$$\Lambda_k^{\text{sc}}(T) = \{\lambda \in \mathbb{C} : \exists Q \in \mathbb{C}^{n \times k}, Q^*Q = I_k, Q^*TQ = \lambda I_k\}.$$

This convention is kept separate from (1). For every such scalar compression,

$$Q^*(\lambda I_n - T)Q = \lambda Q^*Q - Q^*TQ = 0,$$

so $\Lambda_k^{\text{sc}}(T) \subseteq \Lambda_k^{\text{src}}(T)$ for $k \geq 1$. The frame q above gives $1/2 \in \Lambda_1^{\text{sc}}(T)$, while the determinant argument excludes $1/2$ even from the larger source-defined set at $k = 2$. Thus the same witness disproves $\Lambda_1^{\text{sc}}(T) \subseteq \Lambda_2^{\text{sc}}(T)$.

The inclusion is stated unconditionally. Our example uses the permitted dimensions $n = 2$, $k = 1$, and $k + 1 = 2$. It refutes this clause; the separate generic-nonemptiness threshold, spectral envelopes, and arbitrary-dimensional compactness are outside the delivered conclusion.

3 Lean verification and reproducibility

The Lean 4.19.0 project uses a pinned Mathlib revision and its actual `Matrix.rank`, defined by the dimension of a linear-map range. In `lean/Main.lean`:

- `statedRange` and `scalarRange` are the two distinct existential sets above. The orthonormal-column equation is proved to induce an actual Euclidean `Isometry` using the Hilbert-space adjoint.
- The concrete frame identities are exact. For every two-column frame, the determinant is $-1/4$; Mathlib’s invertibility and rank theorems then give rank two. The normalization uses the positive real square root.
- The final theorems negate both inclusions and the universal source-defined inclusion in dimension two, using the verified Hermitian matrix and membership/exclusion pair.

Run these commands inside `lean/`:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

The supplementary `verify.py` checks the witness using exact fractions and expands the general determinant identity as a polynomial in eight independent matrix-entry variables. Fresh-build logs, the axiom audit, and hashes are in `verification/`. The bilingual source is preserved byte for byte in `SOURCE.md`.

References

- [1] The Last Math Competition, conjecture 00000008586, repository statement.
- [2] M.-D. Choi, D. W. Kribs, and K. Życzkowski, *Higher-Rank Numerical Ranges and Compression Problems*, Linear Algebra Appl. 418 (2006), 828–839. arXiv:math/0511278, definition (1).